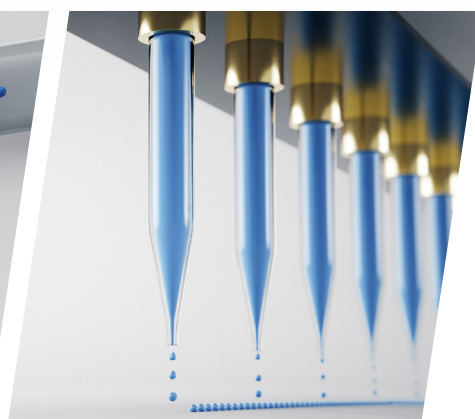
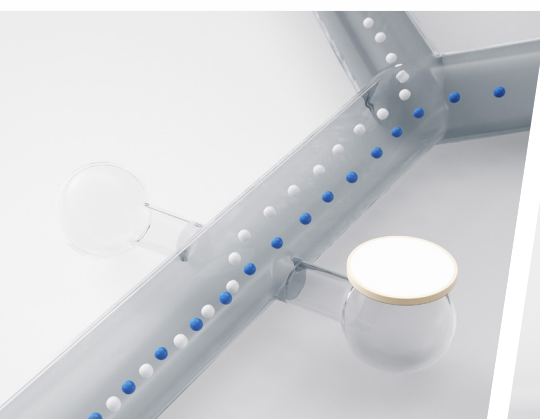
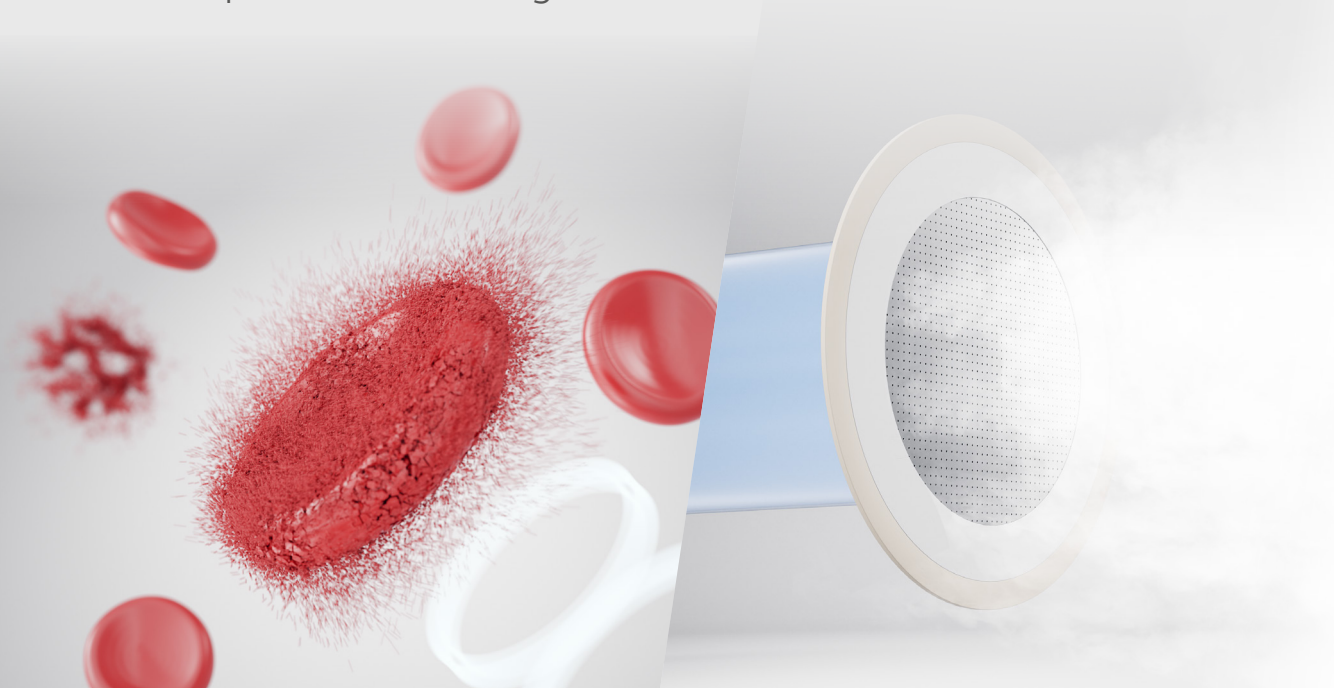


# Piezo Technology for Microfluidics

Effectively Using Piezo Components  
in Medical Technology to Gain  
Competitive Advantage



|                                                                               |           |
|-------------------------------------------------------------------------------|-----------|
| <b>1 Introduction</b>                                                         | <b>3</b>  |
| <b>2 Piezo Technology for Microfluidics</b>                                   | <b>3</b>  |
| 2.1 The Piezoelectric Effect                                                  | 3         |
| 2.2 Piezoceramic Components for Microfluidics                                 | 4         |
| 2.2.1 Materials                                                               | 4         |
| 2.2.2 Designs                                                                 | 4         |
| 2.2.3 Manufacturing technologies                                              | 4         |
| 2.3 Quasistatic and Resonant Mode                                             | 5         |
| 2.4 Unique Selling Points and Advantages of Piezo Technology in Microfluidics | 5         |
| 2.5 Limitations of Piezo Technology in Microfluidics                          | 6         |
| <b>3 Microfluidic Applications in Medical Technology</b>                      | <b>7</b>  |
| 3.1 Mixing, Dispersion, and Cell Lysis with Power Ultrasonic Transducers      | 7         |
| 3.2 Aerosol Generation Using Piezoelectric Membrane Nebulizers                | 8         |
| 3.3 Precise Droplet Generation with Piezoceramic Tube Capillaries             | 9         |
| 3.4 Piezoelectric and Acoustophoretic Single-Cell Sorting                     | 9         |
| <b>4 Summary and Outlook</b>                                                  | <b>10</b> |
| <b>References</b>                                                             | <b>11</b> |
| <b>PI Ceramic in Brief</b>                                                    | <b>12</b> |

## 1 Introduction

The field of microfluidics covers the behavior and manipulation of very small quantities of liquid and gas in the range of nanoliters and picoliters down to femtoliters [1,2]. The fluids are moved, mixed, separated, or otherwise processed in sub-millimeter-sized channels and chambers (Fig. 1).

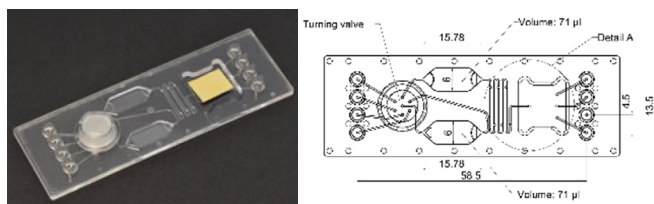


Fig. 1: Active microfluidic platform [3]

The behavior of fluids at these scales is substantially different than in macroscopic systems due to the dominance of effects that can often be disregarded in conventional fluid mechanics [3].

Fundamental to active microfluidics is that much smaller volumes are actively moved than in e.g., conventional chemical or pharmaceutical industrial plants. It is therefore important not to transfer the conventional drive principles for pumps, mixers, and valves 1:1 to the smaller dimensions, but rather to consider alternative, unconventional drives. This enables the development of all-new functions in terms of precision, dynamics, efficiency, parallelism, and mobility. This paper presents piezo technology as a potential unconventional drive technology.

Based on experiences from numerous successfully implemented projects for active microfluidic products, the unique selling points of the technology are highlighted: When does it make sense to use piezo drives?

Knowledge of the limits of the technology is, however, at least just as important for understanding the principles involved: When isn't it worthwhile to use piezo drives, or how can we circumvent undesirable properties?

Microfluidics has revolutionized a number of methods and applications in medical technology. These include in vitro diagnostics in point-of-care devices, various forms of drug delivery into the body, as well as single-cell sorting methods. Consequently, the final part of the white paper will examine the claims about the technology by taking as examples four successful medical technology applications in which piezo actuator technology is the key functional element for the microfluidic solution.

## 2 Piezo Technology for Microfluidics

### 2.1 The Piezoelectric Effect

In certain dielectric solid-state materials, the ions in the crystal lattice move spontaneously when below the Curie temperature, so that the positive and negative center of charge within the unit cells are separated from each other. This phenomenon is also referred to as spontaneous polarization (Fig. 2). The materials are piezoelectric in this state.

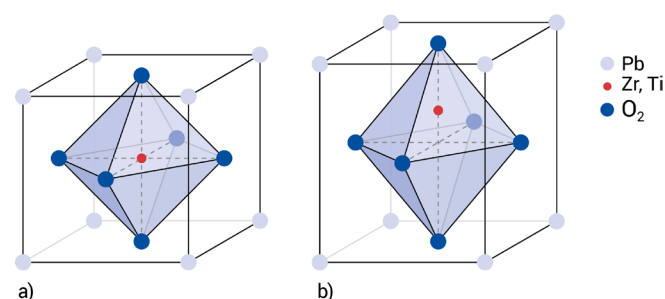


Fig. 2: The piezoelectric effect in lead zirconate titanate (PZT): 1) Unit cell in the symmetrical cubic phase above the Curie temperature; 2) Tetragonal phase with spontaneous polarization below the Curie temperature with piezoelectric behavior

If a metal electrode is attached to a piezoelectric material (Fig. 3) and a voltage is applied in the direction of spontaneous polarization, an internal electric field is created, resulting in a shift of the ions and a change in length. Typical relative changes in length with this actuator effect are approximately 0.1%.

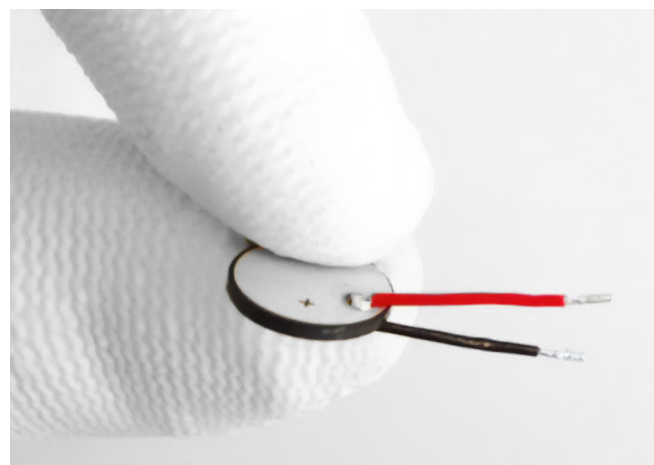


Fig. 3: Piezoceramic disk with metal electrode to generate piezoelectric changes in thickness or to detect pressure signals

Conversely, the exertion of mechanical pressure on the material generates electric charge signals at the metal electrode. This sensor effect is sensitive enough to detect the smallest changes in pressure of ultrasonic signals in the air or in liquids.

## 2.2 Piezoceramic Components for Microfluidics

### 2.2.1 Materials

Active microfluidics requires actuator functions. Polycrystalline piezoceramics have become preferred over piezoelectric single crystals as technically suitable materials in these applications [4]. Ceramics are much simpler to manufacture in different geometric shapes and are more mechanically robust than single crystals (Fig. 3). Lead zirconate titanate (PZT) has established itself among all manufacturers as the material system of choice in this context (Fig. 2). This material system exhibits high piezoelectric coefficients. Its properties can be modified relatively easily by changing the zirconium-titanium ratio or through additives, enabling it to be tailored to the requirements of the application.

The EU's RoHS Directive includes an exemption for the use of lead in piezoceramics [5]. A great deal of research has been devoted to lead-free alternatives for over 20 years. To date, the alternative material systems BNT (bismuth sodium titanate) and KNN (potassium sodium niobate) studied for actuator applications do not achieve the performance of PZT materials; hence, the exemption has been continually extended.

### 2.2.2 Designs

The designs of piezoceramic actuator components for microfluidics can be divided into three categories: longitudinal actuators, bending actuators, and tube actuators (Fig. 4).

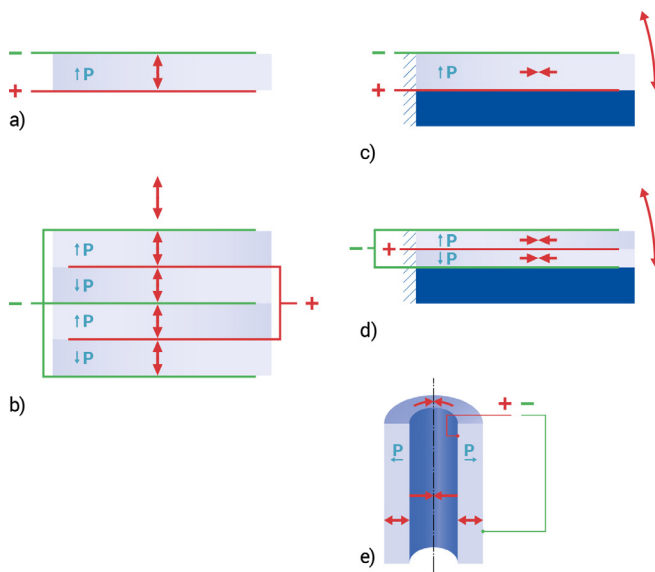


Fig. 4: Designs of piezo actuator components for microfluidics: a) Longitudinal, single-layer; b) Longitudinal, multilayer stacked; c) Bending actuator, single-layer, clamped on one side; d) Bending actuator, multilayer, clamped on one side; e) Tube actuator

Longitudinal actuators act in the direction of spontaneous polarization or the metal electrodes applied, as described in Section 2.1. To increase the achievable displacement, the individual layers can be stacked on top of each other, thereby adding together the displacements of the individual disks.

Bending actuators exploit the characteristic that piezoceramics contract in all directions transverse to the direction of spontaneous polarization at the same time as the longitudinal expansion. This contraction can be transformed into a bending motion if the active element is adhered to a passive substrate and driven. Bending actuators can either be clamped on one side to achieve maximum displacements per length or mounted at both ends to increase the force. Disks with membranes can also be adhered, e.g., for pump functions. Active contraction layers in bending actuators can also have a multilayer design, e.g., to reduce the drive voltages at a constant height.

In ceramic tube actuators, spontaneous polarization is oriented radially. The metal electrodes are accordingly located on the lateral surfaces. When driven, the inner diameter is reduced due to both the contraction of the tube circumference and the change in wall thickness. With thin-walled tubes, the first effect dominates; with thick-walled tubes, the second.

### 2.2.3 Manufacturing technologies

Pressing technology, tape or multilayer technology, thick-film technology, and thin-film or MEMS technology are available to manufacture piezoceramic components for active microfluidics.

In conventional pressing technology, molding involves pressing ceramic powders and then sintering. Typical thicknesses of these bulk ceramic components are in the range of 0.2 to 1.0 mm (Fig. 3).

Tape or multilayer technology supports thinner layers and, in turn, lower drive voltages. The ceramic powder is not pressed, but rather mixed in a polymer binder system and casted into tapes. These tapes are either sintered individually to create substrates with layer thicknesses of at least 40  $\mu\text{m}$  or are printed before sintering with the future electrode material, stacked to create multilayer laminates, and sintered in a co-firing process (Fig. 5). The layer thicknesses within the components are in the range of 20 to 100  $\mu\text{m}$  so that low nominal drive voltages of 10 to 150 V are required to generate actuator functions.

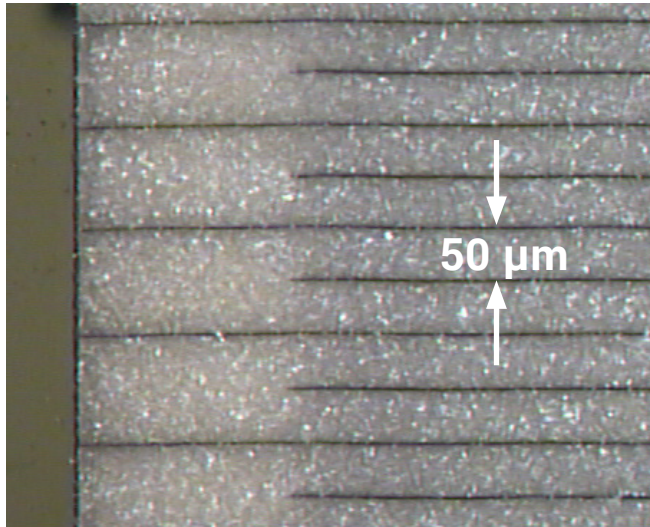


Fig. 5: Piezoceramic multilayer element (sectional view) [PI Ceramic]

Piezoceramic thick films with thicknesses of 10 to 40  $\mu\text{m}$  can also be printed directly onto various substrates (Fig. 6) [6] or applied using aerosol deposition [7]. The aerosol deposition method, wherein particles with high mechanical energy strike surfaces, compact, and undergo thermal treatment, has so far not seen widespread commercial uptake. Screen printing usually requires complex and time-consuming multiple prints. The substrate materials must also withstand the subsequent firing process of at least 900°C. Due to technological requirements, the piezo actuator key figures of the layers are lower than the key figures for bulk or multilayer components, which can for example be adhered to substrates. This procedure is therefore reserved for special applications where, e.g., many elements with small lateral dimensions can be manufactured in parallel or polymer adhesives must be avoided.

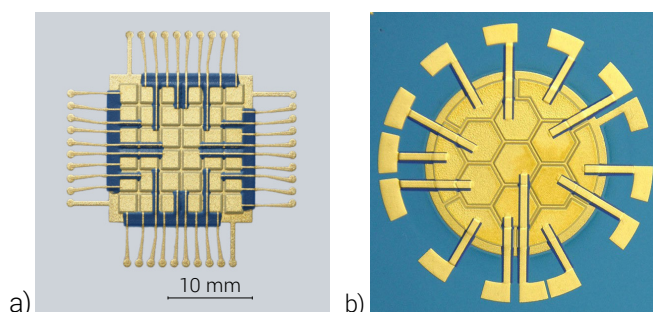


Fig. 6: Examples of piezoceramic thick-film components a) On aluminum oxide substrate [8], b) Using LTCC technology (low temperature co-fired ceramic) [9]

Similar restrictions occur with microelectromechanical systems (MEMS) with piezoelectric thin films (Fig. 7). Layers with thicknesses < 10  $\mu\text{m}$  are applied to silicon substrates using different technologies. High technological barriers to entry, long development times, and the particular challenges in terms of homogeneity and reliability of thin films have thus far lim-

ited this technology to mass-produced applications or multi-channel applications with very small lateral dimensions [10].

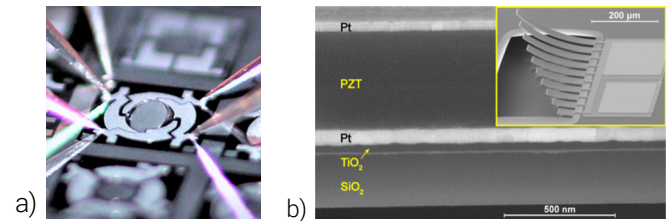


Fig. 7: Examples of piezoelectric thin-film MEMS with active layers made of a) Aluminum nitride (AlN) [11] and b) PZT [12]

## 2.3 Quasistatic and Resonant Mode

With the use of piezo actuator components, a distinction can be drawn between the quasistatic mode below the initial resonant frequency of the overall system and the mode in a resonant frequency of the system.

Quasistatic mode enables the control of defined slew rates, individual cycles, and analog intermediate values, e.g., when opening valves. The term "quasistatic" is in some kind misleading: The high stiffness of piezoceramic components enables microfluidic designs with very high system resonant frequencies so that controlled slew and cycle times down to a few dozen microseconds can be achieved.

In resonant mode, the system is continuously electrically driven with the mechanical resonant frequency in the system. Through the high mechanical quality factors of piezoceramics specifically designed for this, which can reach values over 2,000, very high displacement amplitudes and mechanical power densities are achieved in this mode at low drive voltages. Since the resonant frequencies are normally in the ultrasonic range above 20 kHz, this operating mode is also termed power ultrasound. Because of the high mechanical quality factor of piezo ceramics for high-power ultrasonic applications, high ringing cycles must be taken into account when operating in burst mode.

## 2.4 Unique Selling Points and Advantages of Piezo Technology in Microfluidics

Piezo technology has the following unique selling points and advantages, which can be realized in microfluidic applications and which are explained below:

- A) Extreme motion dynamics
- B) Long lifetime
- C) High precision
- D) Compact design
- E) Bidirectional effect
- F) Energy-efficient operation
- G) No influence of magnetic fields

### A) Extreme motion dynamics

Piezoelectric solutions are nearly indispensable wherever exceptionally dynamic actuator technology is required. The high stiffness of piezoceramic materials and high system resonant frequencies facilitate ultrasonic signals up to the gigahertz frequency range and very low rise times of a few dozen microseconds with mechanical displacements of several dozen micrometers. With these highly dynamic motions, the droplet shape can be controlled, e.g., in inkjet and dispensing processes to prevent uncontrolled satellite droplets. In dispensing systems for highly viscous adhesives, repetition rates of more than 1,000 drops per second are possible. With minute fluid quantities in the picoliter range in inkjet printers, even rates as high as 50,000 drops per second are possible. Droplet speeds in the inkjet process are several meters per second.

### B) Long lifetime

The movement of piezo actuators is based on crystalline solid-state effects, which are not weakened given the right system configuration. Microfluidic designs with piezo actuators normally have no friction couplings or wear processes. This enables system lifetimes of  $10^{11}$  to  $10^{12}$  cycles.

With static and dynamic driving of the actuators, the lifetime depends on the amplitude of the electrical drive field. With AC loads, high fields can cause cracks to develop in the ceramic and the connection of metal electrodes to be compromised. In DC mode, electrochemical degradation processes, which can be accelerated through moisture, dominate at high field strengths.

### C) High precision

The piezoelectric solid-state conversion without friction couplings and the resulting Stick-Slip-Effects can be performed with unlimited resolution. The precision and repeatability of the actuations largely depend solely on the quality of the electronics and sensor technology used. Repeatability in the picometer range can be achieved in piezo-based positioning systems. In inkjet printers, single droplets with volumes <10 picoliters are dispensed continuously and with repeat accuracy.

### D) Compact design

The mechanical design of the solid-state transducers is incredibly simple and compact: a piezoceramic component with typical dimensions in the millimeter range with a thin-film or thick-film metallization with thicknesses in the micrometer range, and electrical wiring.

The use of bending elements that provide displacement transverse to the longest dimension of the components supports a very low structure height or cross-section height of < 1 mm.

### E) Bidirectional effect

The piezoelectric transducer material can be used for both actuator and sensor functions. In this way, very simple closed control loops without additional sensors can be set up to provide specific feedback, e.g., for the reaction to pressure pulses in microfluidic systems. Piezoceramic components are less suited for DC absolute measurements of mechanical parameters due to nonlinearities and limited internal resistances.

### F) Energy-efficient operation

Piezoceramic actuator elements can be compared with electrical capacitors in terms of their energy-related behavior. When driven, they are "charged" with electrical reactive energy. In static mode, they therefore consume almost no energy. During discharge, the electrical reactive energy can be "recovered" with suitable electronics. The necessary circuits are indeed unsuitable for ultraprecise motion. However, motion dynamics are paramount in many microfluidic applications, so electronics with energy recovery can be used. Application in wireless, battery-operated applications or in wearables is entirely possible with this recovery technology.

### G) No influence of magnetic fields

Piezo ceramic actuators can be manufactured with no ferromagnetic materials and do not generate residual magnetic fields during operation. They are therefore very suitable for use under the influence of strong magnetic fields, e.g., in MRI systems, and can be used with no problems in analytics instruments that use magnetic measurement principles.

## 2.5 Limitations of Piezo Technology in Microfluidics

Piezo technology has the following limitations. In some cases, the limitation prevents the technology from being used in microfluidic applications. In most use cases, users must be aware of these limitations to prevent errors when using the technology:

H) Small volumes and pump quantities

I) High electric fields

J) Nonlinear behavior subject to hysteresis

K) Temperature limitation

The individual items are explained in detail below.

### H) Small volumes and pump quantities

The displacements of piezoceramic components have very small amplitudes. They range from a few hundred micrometers with bending elements down to nanometers with single-layer longitudinal actuators (see Section 2.2.2). As such, it is usually not worthwhile to deploy the technology when high flow rates > 100 mL/min, or droplet sizes above the range of a few microliters are needed.

### I) High electric fields

Paragraph B) relating to the lifetime already explained the relationship between the nominal electrical field in the material and the lifetime. Since the achievable displacement amplitudes in piezoceramic components are very small, there is always a trade-off between performance parameters, or the level of the drive field, and the lifetime. Normally the functionally necessary nominal fields are in the range of 0.5 to 3.0 kV/mm, with very thin layers above this in some cases. When designing products, it is advisable to provide only the absolutely necessary field strength in the design and to demonstrate the lifetime through limit tests with excessive values.

### J) Nonlinear behavior subject to hysteresis

Applying high nominal electric fields (see Section I) excites non-linear and hysteresis-related parts of the characteristic curves in the piezoceramic components. With precision applications, it can therefore be necessary to work with sensors and closed control loops for linearization purposes.

Control is not possible in highly dynamic applications above a repetition rate of several hundred hertz. Here the approach either uses linearization models in the drive or transfers from voltage control to charge control, which is, however, unsuitable for static operation.

Hysteresis processes in the material generate active energy, which causes self-heating of the components at high frequencies. This is often not critical in fluidics, as the medium dissipates thermal energy with each cycle. However, self-heating can become the limiting factor in designs with thermal separation of the actuator unit from the fluid, with high field strengths and repetition rates, or in power ultrasound transducers.

### K) Temperature limitation

The piezoelectric effect in piezoceramic components can only take place below the Curie temperature (see Section 2.1). In addition, it is advisable to set the maximum permitted temperature in the component much lower than the Curie temperature to prevent nonlinearities at the phase boundary as well as higher conductivities with resulting degradation processes in the ceramic. A design using a maximum of half the

Curie temperature, expressed in degrees Celsius, is a reasonable limit, based on experience.

As the piezoelectric actuator performance in PZT ceramics with a low Curie temperature in the range of 150 to 200°C is generally better than in ceramics with a high Curie temperature of around 350°C, this empirical value can prove to be limiting or must be taken into account when selecting the material.

## 3 Microfluidic Applications in Medical Technology

Piezo technology is used successfully in very different active microfluidic applications in medical technology [13,14,15]. These include the delivery of liquid analgetics with piezoelectric valves from implanted reservoirs directly into the spinal canal [16], reliable sample preparation and fluid handling within point-of-care analyzers in in-vitro diagnostics [17], selection and application of single cells or spheroids on microarrays in high-throughput experiments [18], implantable micropumps to regulate intraocular pressure [19], and acoustofluidic manipulation of complete cell clusters to produce defined tissue structures [20].

Four example applications of piezo technology in active microfluidics are presented in more detail in the following sections.

### 3.1 Mixing, Dispersion, and Cell Lysis with Power Ultrasonic Transducers

Turbulence-free laminar flows tend to occur in the narrow channels of microfluidic systems so that a defined blending of two or more substances or an even dispersion of particles in a fluid barely occurs with normal flows. That is why microfluidic chips require passive or active mixing functions [21].



Fig. 8: Miniaturized 35 kHz power ultrasonic transducer for reliable mixing, dispersion, and cell lysis in microfluidic systems (PI Ceramic)

An active solution for reliable mixing or dispersion consists of introducing ultrasound [22] from a miniaturized power ultrasonic transducer (Fig. 8), which acts on a reaction chamber

of the chip in resonance or burst mode with its ultrasonic horn (Fig. 9).

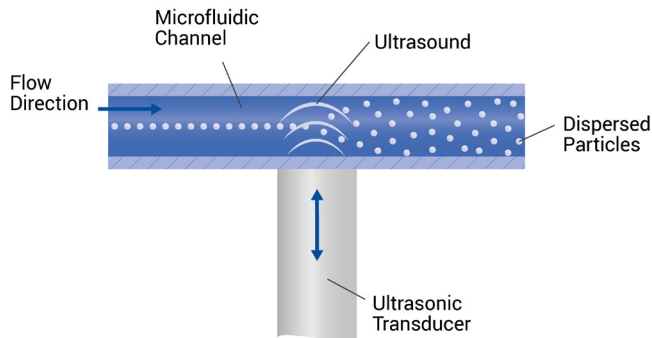


Fig. 9: Dispersion of particles in a microfluidic system using power ultrasound [23]

In the preparation of microfluidic blood samples, increasing the ultrasonic power in the channel causes cell lysis, i.e., the rupture of blood cells, due to the cavitation introduced [24]. Cell components such as DNA can then be analyzed by binding them to microbeads, i.e., magnetic particles dispersed by ultrasound. In a functional combination, the sample is heated simultaneously throughout the channel cross-section by the ultrasound to the required analysis temperature of approximately 50°C [25].

## 3.2 Aerosol Generation Using Piezoelectric Membrane Nebulizers

To treat respiratory diseases, drugs are administered as respirable aerosols. Among the various techniques known for aerosol generation [26], active piezoelectric mesh nebulizers have become the first choice in recent years due to their efficiency, portability, and short response times [27,28].

The structure of the functional element consists of a thin piezoceramic ring with an outer diameter of approximately 15 to 20 mm. This ring causes a thin perforated stainless-steel membrane to vibrate as it moves in the inner diameter. The membrane is approximately 50 µm thick and has several thousand laser-machined, slightly conical holes with diameters in the range of a few micrometers (Fig. 10).

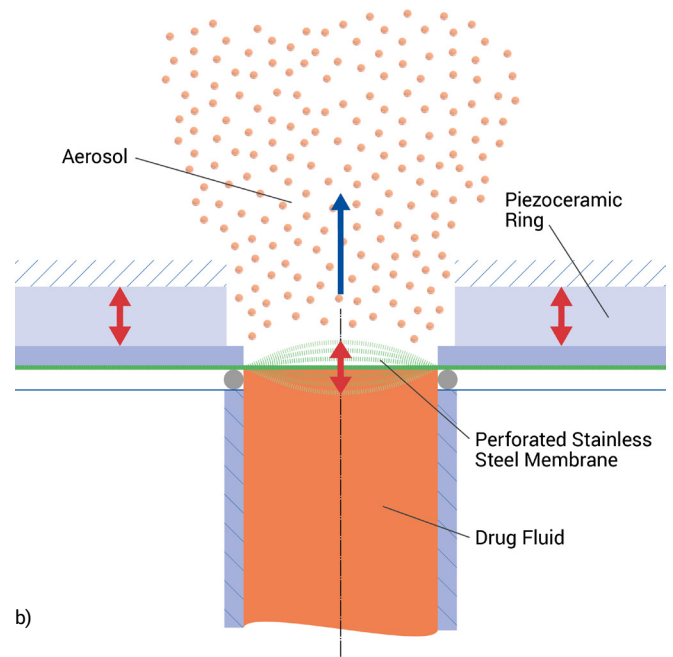
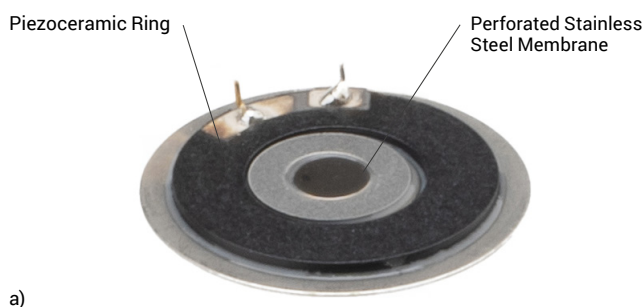


Fig. 10: Piezoelectric membrane nebulizer: a) Typical design; b) Functional sketch

The drive frequency of the ring element is matched to the basic bending resonance frequency of the membrane, which is approximately 100 to 200 kHz depending on the system. The drive voltages are in the range of a few dozen volts. The system operates in burst mode.

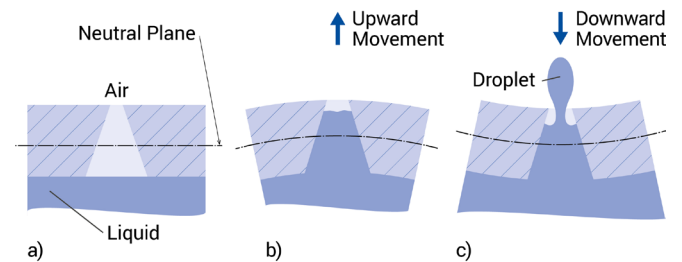


Fig. 11: Droplet formation on the perforated membrane of a nebulizer during a vibration cycle: a) Idle state – surface tension retains fluid; b) Loading of the fluid into the nozzle; c) Expulsion of the droplet during the backswing (figure based on [29])

Fig. 11 explains the generation of aerosol droplets with dimensions of a few micrometers. The materials used, hole geometries, and droplet sizes are matched to the specifications of the particular drugs, e.g., their viscosity.

Smart drive systems respond to the patient's breathing rhythm, thus maximizing the efficiency of the administered quantity of drug. Typical nebulization quantities are around several milliliters per minute.

### 3.3 Precise Droplet Generation with Piezoceramic Tube Capillaries

The precise, contamination-free, and rapid generation of individual fluid droplets with repeatable volumes in the picoliter range is required in many areas of medical technology. These microdroplets are needed, e.g., to develop new drugs with high-throughput screening, for the manufacture of patient-specific drugs, or in sensitive analysis assays with defined reagent quantities [30,31,32].

A technology that is ideally suited for the said tasks is jet dispensing using piezoceramic tube capillaries [33].

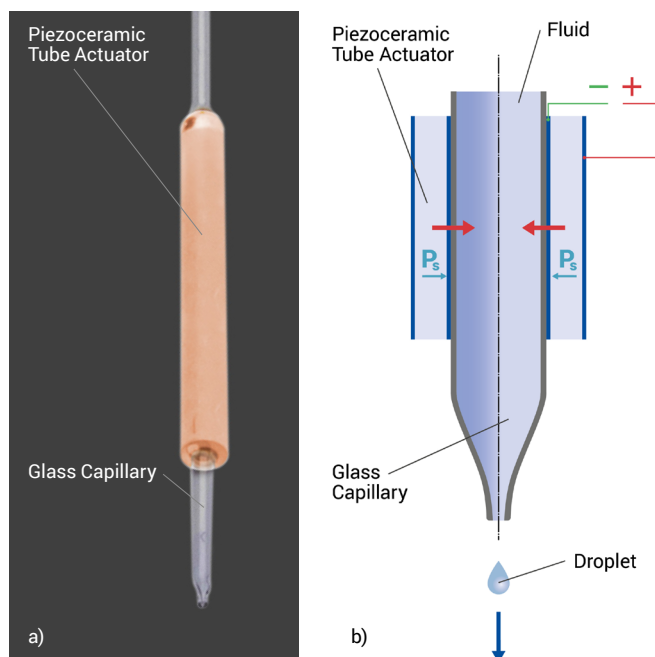


Fig. 12: Glass capillary with glued piezoceramic tube actuator for the contactless generation of precise individual droplets: a) Typical design; b) Functional sketch

Fig. 12 shows a typical design and the operating principle: A piezoceramic tube actuator with an outer diameter of approximately 1 to 2 mm and a length of 12 to 20 mm is adhered to a standard glass capillary tube. Driving the actuator electrodes with voltages of approximately 100 volts and pulse widths in the range of 50  $\mu$ s creates a dynamic contraction signal with a defined time progression. This triggers a pressure wave in the fluid, which overcomes the surface tension at the tip of the capillary and releases a droplet with a volume of several hundred picoliters. This jet droplet then travels several millimeters to the target position.

The fluid touches only the inner surfaces of the capillary tube so that the system is very easy to clean and enables contamination-free processing. These inner surfaces can also be provided with, e.g., hydrophilic or hydrophobic functional coat-

ings, depending on the reagent, to further improve the repeatability of droplet formation.

Combining piezoceramic capillary tubes with a camera, stroboscopic technology, and a three-axis portal system makes them excellently suited for the aforementioned tasks.

The system can also be used to place individual cells or cell clusters into microarrays in a defined manner. This involves image recognition that checks whether only one object is in the droplet, wherein droplets with multiple objects are discarded into a waste reservoir.

### 3.4 Piezoelectric and Acoustophoretic Single-Cell Sorting

In many medical research areas and therapies, e.g., stem cell research or blood cancer therapy, individual cells need to be separated. Therefore, flow cytometers for permanent cell analysis in the microfluidic channel are coupled with an adjacent sorting function. Various processes are available for this sorting [34,35]. Two piezoelectric methods are presented below.

Fig. 13 highlights how particles or cells can be sorted with highly dynamic pressure pulses generated piezoelectrically from bending membranes [36,37,38].

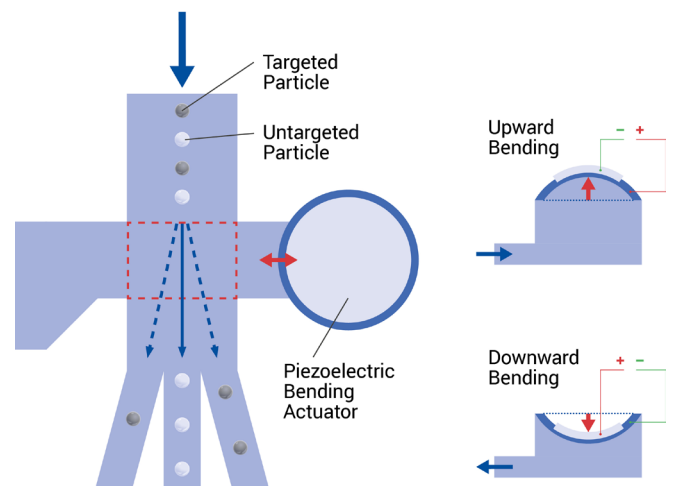


Fig. 13: Schematic representation of cell sorting of cytometrically marked cells with piezoelectrically generated dynamic pressure pulses (figure based on [36])

There are systems that perform therapeutic sorting tasks using piezoelectric actuators in 24 channels in parallel to achieve sorting rates of 180 million cells per hour [39]. This corresponds to a controlled repetition rate of 2 kHz.

Where cells or carrier particles differ in terms of size, density, or compressibility, they can be processed using acoustophoretic sorting [40,41] (Fig. 14).

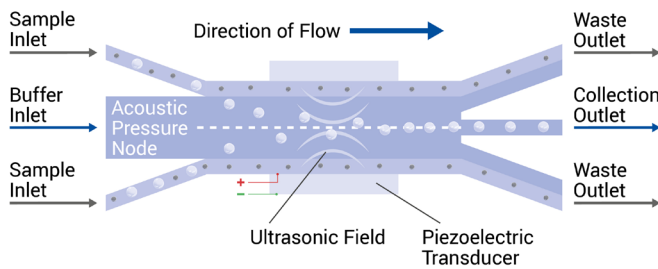


Fig. 14: Schematic representation of permanent acoustophoretic cell sorting by standing ultrasonic waves without cytometry and marking (figure based on [42])

The illustrated process does not require cytometry and single-cell marking since the objects are separated from each other only through their physical characteristics. The cells respond differently to standing acoustic waves and their nodes and antinodes that are generated by piezoelectric transducers in resonant mode. This enables very high sorting rates to be achieved, as permanent separation occurs in the flow without explicit actuating signals.

## 4 Summary and Outlook

Piezo technology is exceptionally suited for fulfilling specific requirements of active microfluidics in wide-ranging medical technology applications.

To use the technology effectively, prospective users must appreciate both the unique selling points and advantages as well as the limitations of piezo actuators in microfluidics. This white paper examined these issues in detail.

The potential of piezo technology for the further development of microfluidics has not yet been fully exploited in many areas. Applications such as single-cell handling with acoustic tweezers and biological tissue synthesis with ultrasound are examples of developments that are still in their early stages.

## References

- [1] Tabeling, P.: Introduction to Microfluidics. Oxford University Press, 2nd Ed., 2023.
- [2] Manz, A., Neuzil, P., O'Connor, J. S., Simone, G.: Microfluidics and Lab-on-a-Chip. The Royal Society of Chemistry, 2020.
- [3] microfluidic ChipShop GmbH, Catalogue 2023, p.146, [www.microfluidic-chipshop.com](http://www.microfluidic-chipshop.com).
- [4] Friend, J., Yeo, L.: Piezoelectric Materials for Microfluidics. In: Li, D. (eds) Encyclopedia of Microfluidics and Nanofluidics. Springer, Boston, MA, 2008.
- [5] Directive 2011/65/EU of the European Parliament and of the Council of 8 June 2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment. Exemption 7©-I, [https://environment.ec.europa.eu/topics/waste-and-recycling/rohs-directive\\_en](https://environment.ec.europa.eu/topics/waste-and-recycling/rohs-directive_en).
- [6] Santo Zarnik, M., Uršič, H., Kosec, M.: Recent progress in thick-film piezoelectric actuators prepared by screen-printing. pp. 1-27, Chapter in book: Piezoelectric Actuators. Nova Science Publishers, New York, 2011, ISBN 978-1-61324-181-3.
- [7] Patil, D.R.; Annapureddy, V.; Kaarthik, J.; Thakre, A.; Akedo, J.; Ryu, J. Piezoelectric Thick Film Deposition via Powder/Granule Spray in Vacuum: A Review. Actuators, Vol. 9 (2020) p. 59, <https://doi.org/10.3390/act9030059>.
- [8] Qiu, Y., Wang, H., Gebhardt, S., Bolhovitins, A., Démoré, C.E.M., Schönecker, A., Cochran, S.: Screen-printed ultrasonic 2-D matrix array transducers for microparticle manipulation, Ultrasonics, Volume 62 (2015) pp. 136-146, <https://doi.org/10.1016/j.ultras.2015.05.010>.
- [9] Bramlage, B., Gebhardt, S.: Development of PZT thick film actuators for active micro-optics. In Proc. ACTUATOR 2010, pp. 594-596.
- [10] Roy, K., Lee, J.E.Y., Lee, C. Thin-film PMUTs: a review of over 40 years of research. Microsyst Nanoeng, Vol. 9 (2023) p. 95, <https://doi.org/10.1038/s41378-023-00555-7>.
- [11] Meinel K, Melzer M, Stoeckel C, Shaporin A, Forke R, Zimmermann S, Hiller K, Otto T, Kuhn H. 2D Scanning Micromirror with Large Scan Angle and Monolithically Integrated Angle Sensors Based on Piezoelectric Thin Film Aluminum Nitride. Sensors. Vol. 22 (2020) p. 6599, <https://doi.org/10.3390/s20226599>.
- [12] Das, D, Sanchez, L, Martin, J, et al. Control of ferroelectric and linear piezoelectric response of PZT films through texture. J Am Ceram Soc., Vol. 101 (2018) pp. 2965–2975, <https://doi.org/10.1111/jace.15419>.
- [13] Liu Y, Yin Q, Luo Y, Huang Z, Cheng Q, Zhang W, Zhou B, Zhou Y, Ma Z. Manipulation with sound and vibration: A review on the micromanipulation system based on sub-MHz acoustic waves. Ultrason Sonochem., Jun. (2023), doi: 10.1016/j.ulsonch.2023.106441.
- [14] Yan, Q.; Yin, Y.; Sun, W.; Fu, J. Advances in Valveless Piezoelectric Pumps. Appl. Sci., Vol. 11 (2021) p. 7061, <https://doi.org/10.3390/app11157061>.
- [15] Asadi Dereshgi, H., Dal, H., Yildiz, M.Z.: Piezoelectric micropumps: state of the art review. Microsyst Technol 27 (2021) pp. 4127–4155, <https://doi.org/10.1007/s00542-020-05190-0>.
- [16] Chappel, E.: Implantable drug delivery devices. Chapter in Book: Drug delivery devices and therapeutic systems. Academic Press, 2021, pp.129-156, 10.1016/B978-0-12-819838-4.00001-8.
- [17] Niemz, A., Ferguson, T., Boyle, D.: Point-of-Care Nucleic Acid Testing for Infectious Diseases. Trends in biotechnology. 29 (2011) pp. 240-50. 10.1016/j.tibtech.2011.01.007.
- [18] Fernandez Vallone, V., Telugu, N., Fischer, I., Miller, D., Schommer, S., Diecke, S., Stachelscheid, H.: Methods for Automated Single Cell Isolation and Sub-Cloning of Human Pluripotent Stem Cells. Current protocols in stem cell biology. 55 (2020). e123, 10.1002/cpsc.123.
- [19] Bußmann, A., Leistner, H., Zhou, D., Wackerle, M., Congar, Y., Richter, M., Hubbuch, J.: Piezoelectric Silicon Micropump for Drug Delivery Applications. Applied Sciences, 11 (2021) p. 8008, 10.3390/app11178008.
- [20] Ma, Z., Melde, K., Athanassiadis, A., Schau, M., Richter, H., Qiu, T., Fischer, P.: Spatial ultrasound modulation by digitally controlling microbubble arrays. Nature Communications. 11. (2020) p. 4537. 10.1038/s41467-020-18347-2.
- [21] Lee, C.-Y., Chang, C.-L., Wang, Y.-N., Fu, L.-M.: Microfluidic Mixing: A Review. Int. J. Mol. Sci., 12 (2011) pp. 3263-3287; doi:10.3390/ijms12053263.
- [22] Lim, E., Lee, L., Yeo, L. Y., Hung, Y. M., Tan, M. K.: Acoustically Driven Micromixing: Effect of Transducer Geometry. in IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control, vol. 66, no. 8 (2019) pp. 1387-1394, doi: 10.1109/TUFFC.2019.2920683.
- [23] Iranmanesh, I., Ohlin, M., Ramachandraiah, H., Ye, S., Russom, A., Wiklund, M.: Acoustic micro-vortexing of fluids, particles and cells in disposable microfluidic chips. Biomedical Microdevices. 18 (2016). 10.1007/s10544-016-0097-4.
- [24] Shehadul Islam, M., Aryasomayajula, A., Selvaganapathy, P.R.: A Review on Macroscale and Microscale Cell Lysis Methods. Micromachines. Vol. 8, No. 3 (2017) p. 83. doi: 10.3390/mi8030083. PMID: PMC6190294.
- [25] Lee, H., Oh, M. T., Lee, Y. J., Son, K. H., Choi, N., Lee, B. C., Lee, S. H.: Highly-efficient microfluidic ultrasonic transducers assisted gDNA extraction system in whole blood for POCT applications, Sensors and Actuators B: Chemical, Volume 319 (2020) p. 128317, <https://doi.org/10.1016/j.snb.2020.128317>.
- [26] Vecellio, L.: The mesh nebuliser: a recent technical innovation for aerosol delivery. Breathe, Vol. 2, No. 3 (2006) pp. 252-260; DOI: 10.1183/18106838.0203.252.
- [27] Ari A.: Jet, Ultrasonic, and Mesh Nebulizers: An Evaluation of Nebulizers for Better Clinical Outcomes. Eurasian Journal of Pulmonology, 16 (2014) pp. 1-7.
- [28] Pritchard, J.N., Hatley, R. H.M., Denyer, J., von Hollen, D.: Mesh nebulizers have become the first choice for new nebulized pharmaceutical drug developments, Therapeutic Delivery, Vol. 9, No. 2, (2018) pp. 121-136.
- [29] Guerra-Bravo, E.; Lee, H.-J.; Baltazar, A.; Loh, K.J.: Vibration Analysis of a Piezoelectric Ultrasonic Atomizer to Control Atomization Rate. Appl. Sci., Vol. 11 (2021) p. 8350. <https://doi.org/10.3390/app11188350>
- [30] Li, H., Liu, J., Li, K., Liu, Y.: Piezoelectric micro-jet devices: A review. Sensors and Actuators A: Physical, Volume 297 (2019) 111552, <https://doi.org/10.1016/j.sna.2019.111552>.
- [31] Zhu, P., Wang, L.: Passive and active droplet generation with microfluidics: a review. Lab Chip, Vol. 17 (2016) pp. 34-75. 10.1039/C6LC01018K.
- [32] Friend, J., Yeo, L.: Piezoelectric Microdispenser. In: Li, D. (eds) Encyclopedia of Microfluidics and Nanofluidics. Springer, Boston, MA, 2008, [https://doi.org/10.1007/978-0-387-48998-8\\_1243](https://doi.org/10.1007/978-0-387-48998-8_1243).
- [33] Zhong, Y., Fang, H., Ma, Q., Dong, X.: Analysis of droplet stability after ejection from an inkjet nozzle. J. Fluid Mech., Vol. 845 (2018) pp. 378–391 doi:10.1017/jfm.2018.251.
- [34] Wyatt Shields, C.; Reyes, C. D., López, G. P.: Microfluidic Cell Sorting: A Review of the Advances in the Separation of Cells from Debunking to Rare Cell Isolation. Lab Chip, Vol. 15, No. 5 (2015) pp. 1230–1249, doi: 10.1039/c4lc01246a.
- [35] Gong, Y., Fan, N., Yang, X., Peng, B., Jiang, H.: New advances in microfluidic flow cytometry. ELECTROPHORESIS, Vol. 40 (2018) 10.1002/elps.201800298.
- [36] Cho, S. H. et al.: Micro-fabricated fluorescence-activated cell sorter. 2009 Annual International Conference of the IEEE Engineering in Medicine and Biology Society (2009) pp. 1075-1078.
- [37] Cho, S., Godin, J., Chen, C.-H., Qiao, W., Lee, H., Lo, Y.-H.: Review Article: Recent advancements in optofluidic flow cytometer. Biomicrofluidics. Vol. 4 (2010) 43001. 10.1063/1.3511706.
- [38] Kraus, D., Kleiber, A., Ehrhardt, E., Leifheit, M., Horbert, P., Urban, M., Gleichmann, N., Mayer, G., Popp, J., Henkel, T.: (2021). Three step flow focusing enables image-based discrimination and sorting of late stage 1 Haematococcus pluvialis cells. Plos one, Vol. 16, No. 3 (2021) e0249192.
- [39] Hulsas, R., Villa-Komaroff, L., Koksai, E., Korsgren, O., Sharpe, J. C., Berglund, D.: Purification of regulatory T cells with the use of a fully enclosed high-speed microfluidic system. Cytotherapy, Vol. 16, No. 10 (2014) pp. 1384-1389, DOI:<https://doi.org/10.1016/j.jcyt.2014.05.016>
- [40] Friend, J., Yeo, L.: Microscale Acoustofluidics: Microfluidics Driven via Acoustics and Ultrasonics. Reviews of Modern Physics, 83 (2011) pp. 647-704. 10.1103/RevModPhys.83.647.
- [41] Connacher, W., Zhang, N., Huang, A., Mei, J., Zhang, S., Gopesh, T., Friend, J.: Micro/nano acoustofluidics: materials, phenomena, design, devices, and applications. Lab Chip, Vol. 18, No. 14 (2018) pp. 1952-1996, DOI: 10.1039/c8lc00112.
- [42] Yang, A. H. J., Soh, H.T.: Acoustophoretic Sorting of Viable Mammalian Cells in a Microfluidic Device. Anal. Chem., Vol. 84, No. 24, (2012) pp. 10756–10762, <https://doi.org/10.1021/ac3026674>.

## PI Ceramic in Brief

Redefining the limits of what can be measured and moved - together with our customers. As a global partner with more than thirty years of expertise, PI Ceramic develops and manufactures sophisticated piezoceramic components and sub-systems for applications in the areas of medical technology, industrial ultrasonic technology, and precision dosing. At the site in Lederhose (Germany), one hundred of the current over 400 employees work in research and development. PI Ceramic is part of the PI Group, the innovation and market leader in high-precision positioning technology.

### Physik Ceramic GmbH

Lindenstraße

D-07589 Lederhose, Germany

[www.piceramic.com](http://www.piceramic.com)